

CONTACTS

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- Timișoara, Romania

EDUCATION

10.2024 — Present (Expected 2027)
West University of Timișoara
Faculty of Math & Computer Science
BSc in Computer Science (English Stream)

LANGUAGES

Romanian Native (C2)
English C2 (Cambridge CAE Grade A, 200+)
French A2 — B1

EXTRACURRICULAR

- Competitive Debate Trainer & Referee**
- Chrome Debate Club** (C.D. Loga National College, Timișoara) — Trainer and tournament referee; coached debaters in formal logic and persuasive argumentation.
 - Solaris Debate Club** (Grigore Moisil High School) — Club member and competitive speaker.
- Volunteering & Community**
- EducaTM:** Refurbished laptops for underprivileged students lacking online school access.
 - District23 Festival:** Event operations volunteer.

Lucas Rus

Software Engineer

EXPERIENCE

02.2026 — 08.2026
Vitas Romania
Software Engineering Intern (Automation & Dev)
Trial Balance Parser & Financial Statement Mapper (02–06.2026): Automated SME credit underwriting by parsing trial balance PDFs (*Balanță de Verificare*) into official Balance Sheets (*Bilanț*) and P&L statements. Deterministic coordinate boundary reconstructor, SHA-256 caching (<2s), and OMFP 1802/2014 account netting down to the cent, with Streamlit underwriting portal.
Python, pdfplumber, OpenCV, OpenAI API (Vision), Streamlit, Pandas, OpenPyXL
Company Data Hub v2 (06–08.2026): Corporate BI platform indexing **1.8M+ entities** and 16 years of open-source fiscal filings. Generates 360° company profiles (administrators, ownership, revenue, solvency) and portfolio tracking with snapshot diffing to monitor changes over time. Powered by a sub-5ms DuckDB OLAP engine with atomic snapshot migrations and 130 Pytest test suites (MIT license).
DuckDB, Python, Parquet, Polars, Pytest (130 tests), SQL, Corporate BI & OLAP

11.2024 — Present
JoinStellar.ai
AI Training & Agent Evaluation Contractor
Train frontier coding agents to autonomously resolve real-world GitHub Pull Requests and generate comprehensive test harnesses for **Grafana** and **Godot**. Benchmark multi-turn agent execution trajectories, stress-testing tool call sequences, code repair loops, and database modification states.

09.2024 — 08.2025
Arhebis Digital Systems
Junior Developer

Automated digital production in Photoshop & Illustrator via ExtendScript JavaScript to batch-process illustrations and typography for book publications, cutting graphic prep time by 60%.
Built Python automation tools interfacing with Google Maps API to geocode addresses and automate POI maps for publications; engineered end-to-end XML/XHTML and CSS transformation pipelines for reflowable ePub3 books.

10.2024 — 07.2025
Outlier & DataAnnotation.tech
AI Model Evaluator & Prompt Engineer

Outlier (05–07.2025): EN-RO prompt localization, prompt repair, code evaluation, safety red-teaming, and RLHF comparative ranking.
DataAnnotation (10–12.2024): 5- to 7-axis model evaluation, PII audits, and code syntax validation.

12.2023 — 03.2024
Buildcorpmat
Web Developer & IT Systems Admin
Revamped website & brand in HTML5/SCSS/JS; built customer mailing system; administered DB and domain infrastructure (@buildcorpmat.ro).

SKILLS

LANGUAGES
Python, JavaScript (ES6+), TypeScript, Java 17+, C++, C#, SQL, HTML5, CSS3/SCSS, XML/XHTML
FRONTEND & UX
React.js (18/19), Next.js (App Router), Tailwind CSS v4, Vite, Bootstrap, Avalonia .NET (C#), SVG Animation, WebSockets (STOMP), Gradio, Streamlit
BACKEND & DATA
FastAPI, Flask, Node.js, Spring Boot 3 (Hibernate JPA, Spring Security), REST APIs, DuckDB (OLAP), PostgreSQL, MySQL, H2, Docker, Linux/Bash
AI & COMPUTER VISION
PyTorch, EfficientNet, Multimodal LLMs (Gemini, OpenAI, OpenRouter), Layout OCR (Docling, Tesseract), Agentic Workflows, RLHF, Prompt Architecture
SECURITY & INFRA
MITRE ATT&CK, Splunk SIEM, Snort IDS/IPS, pfSense Firewalls, Okta IAM, OWASP Top 10, Supabase (RLS & Triggers), DNS Management

SYSTEMS & TOOLS
Win32 APIs, Cocoa/iOSKit, Git/GitHub, Unity C#, Photoshop, Illustrator, ExtendScript JS, ePub3

CERTIFICATIONS

Google.org Cybersecurity Program
Virtual Routes at UVVT (04–08.2025)
Hands-on labs: MITRE ATT&CK, AI SecOps, Splunk SIEM/SOC, Snort IDS, pfSense, Okta IAM, OWASP Top 10, IT Audit.

Cambridge English: Advanced (CAE)
Grade A — Score 200+ (Certified CEFR C2)
Highest grade awarded; certified bilingual / native English equivalence.

Professional Competence in CS
Romanian Ministry of Education (2024)
Intensive Math & Informatics track. Capstone: *Ludo* (3D chemical synthesis engine in Unity C#).

IC3 Digital Literacy Certification
Computing Fundamentals & Networks

HACKATHONS & CTF

iTec National Hackathon — 3rd Place
Game Development / Unity C# (Future Jump)

Infoeducatie Olympiad — 2nd Place
National Computer Science Olympiad

Competitive Cybersecurity CTFs
UVT CTF Team Competitor
Active participant in **iTec 2025 CTF**, **PicoCTF**, and **UniVsThreats CTF** (web exploitation, cryptography, network forensics).

CORE PHILOSOPHIES

- **Deterministic AI:** Pairing stochastic LLMs with strict schemas, compiler feedback loops, and mathematical parity.
- **High-Velocity Prototyping:** Compressing execution cycles from weeks to days with agentic workflows.
- **Tactile UX:** Obsession with fluid transitions, sub-5ms data responses, and unconventional visual maps.

FEATURED ENGINEERING & AI PROJECTS

GigTim
Upcoming Flagship Venture (In Dev) | gigtim.arhebis.ro
Next-generation HoReCa day-laborer marketplace (*zillieri*, Legea 52/2011) designed to modernize short-term staffing and transform local employment. Replaces informal hiring with instant shift discovery, interactive animated SVG map of Timișoara with live route paths, automated ITM compliance CSV generation, and deferred GDPR identity verification.
Next.js 16 (React 19), Tailwind CSS v4, Supabase (PostgreSQL, Realtime)

Economic Collapse Forecast
In Active Development | Sovereign Debt EWS
Sovereign crisis Early Warning System (in active development) predicting debt defaults, currency collapses, and banking crises 2 years in advance across 261 countries. Dual-engine: Variable Selection Networks (VSN) + LSTM Autoencoders paired with attention classifier and GDELT sentiment (**75.4% precision, 0.947 AUC**).
PyTorch, Temporal Fusion Transformers (TFT), GDELT API, Polars, Scikit-learn

ContaAI
github.com/lucas-rus/Conta
Automated document OCR and audit engine pairing a FastAPI backend with an **Avalonia .NET C# desktop UI** for macOS/Windows. Integrates multi-provider LLMs (Gemini, OpenAI, OpenRouter) with a 6-axis confidence score, live Romanian ANAF API CUI validation, and e-Factura/SAGA XML export.
FastAPI, Python, Avalonia .NET (C#), ANAF API, Gemini / OpenAI

Algorithmic Studies & CSP
github.com/lucas-rus/sat_algo-benchmark
Comparative SAT Solving: Benchmarked Resolution, DP, DPLL, and Glucose3 (via PySAT) measuring memory and runtime bottlenecks with psutil and Matplotlib.
AC-2001 Algorithm: Implemented Bessiere et al.'s optimal Arc Consistency in C++ for the Map Coloring problem using support pointers to achieve optimal O(ed²) complexity.
Python, PySAT, psutil, Matplotlib, C++11 (ac2001_algo)

CoFound
1st Place Winner | cofound-app.vercel.app
Startup collaboration platform for equity/side-projects. Won **1st Place at UVT Prototype Fair 2025**. Built responsive React 18 frontend with STOMP WebSockets for real-time messaging, application tracking, skill-matching search, and peer reviews.
React 18, Vite, Bootstrap, Java Spring Boot 3, Hibernate JPA, MySQL, WebSockets

German Legal Book Digitizer
Enterprise Pipeline (with Arhebis) | 400k Pages
Digitization pipeline built in collaboration with **Arhebis Digital Systems** for German legal commentaries, **improving turnaround time by ~75% and cutting costs by 90%** over legacy workflows. Engineered an **agentic self-correction loop** validating LLM outputs against strict DTD schemas (jurkommentar.dtd) with auto-repair (Docling CUDA, Tesseract OCR).
Python, Docling (CUDA), Tesseract OCR, Gemini API, lxml, Streamlit

Font Recognition AI
github.com/lucas-rus/font-recognition-ai
26-class print font family classifier using EfficientNet-B0 (4.04M params) trained on 312k synthetic documents. Achieved **93.82% Top-1** and **96.53% Top-3** accuracy. Built interactive Gradio UI featuring multi-scale sliding-window patch extraction and ink-density-weighted probability aggregation.
PyTorch, EfficientNet-B0, OpenCV, Gradio UI, Synthetic Data Synthesis

Systems & Creative Tech
Native C++ & Unity 3D
Mouse Mover: Ultra-lightweight native OS background utility in C++11/Obj-C++ using direct Win32 SendInput and macOS Cocoa/iOS APIs.
Ludo & Future Jump: *Ludo* is an educational 3D chemistry simulation & synthesis game in Unity C# modeling chemical reactions and commercial sales loops (CS Capstone). *Future Jump* won 3rd Place at iTec Hackathon.
C++11, Objective-C++, Win32 APIs, Cocoa/iOS, Unity C# (lucas-rus.itch.io)